Bilayer square lattice

Phase diagram from a neural quantum state

1 Hamiltonian

The lattice is a bilayer square lattice. The index i runs over the lattice sites of the square lattice, and $\ell = 1, 2$ is the layer. The fermion $c_{i\ell\sigma}$ carries spin $\sigma \in \{\uparrow, \downarrow\}$.

The spin on layer ℓ of lattice site i is

$$\boldsymbol{S}_{i\ell} = \frac{1}{2} \sum_{\sigma\sigma'} c^\dagger_{i\ell\sigma}\, \boldsymbol{\sigma}_{\sigma\sigma'}\, c_{i\ell\sigma'}. \tag{1}$$

Nearest-neighbor pairs on one layer are $\langle ij \rangle$. Next-nearest-neighbor pairs, the diagonals of the plaquettes, are $\langle\langle ij \rangle\rangle$. The Hamiltonian is

$$\begin{aligned} H = & -t_1 \sum_{\langle ij\rangle,\ell,\sigma} \left(c^\dagger_{i\ell\sigma} c_{j\ell\sigma} + \text{h.c.}\right) - t_2 \sum_{\langle\langle ij\rangle\rangle,\ell,\sigma} \left(c^\dagger_{i\ell\sigma} c_{j\ell\sigma} + \text{h.c.}\right) \\ & + J \sum_i \boldsymbol{S}_{i1} \cdot \boldsymbol{S}_{i2}. \end{aligned} \tag{2}$$

There is no interlayer hopping. $J > 0$ is antiferromagnetic. Set $t_1 = 1$. This fixes the energy unit. Set $t_2 = 0.2$.

Each lattice site has two layers and two spin states, four single-particle states in all. The filling ν is the number of electrons divided by four times the number of lattice sites. Equivalently, ν is the electron number per layer, per spin, divided by the number of lattice sites. Full filling is $\nu = 1$: both layers and both spins are occupied. Half filling is $\nu = 1/2$. Quarter filling is $\nu = 1/4$. The particles need not be polarized; ν fixes only the density.

2 Task

Solve H by variational Monte Carlo with a neural-network wavefunction. State the lattice size, the boundary conditions, the ansatz, the sampler, the optimizer, and the statistical uncertainty on every reported number.

2.1 Line cuts

Compute two cuts through the ground-state phase diagram, each against J, from weak coupling through strong coupling. One cut is at half filling, $\nu = 1/2$. The other is at quarter filling, $\nu = 1/4$. For each point, report the energy per lattice site and the diagnostics that decide the phase: a charge gap, a spin gap, a magnetic order parameter, a superconducting pair correlation, and the momentum distribution. Say whether that point is a Fermi liquid, a gapped state that preserves every symmetry of H, a superconductor, or another phase. A label without the diagnostic is not an answer.

On the half-filling cut, identify whether a gapped phase is present.

2.2 Phase diagram

Repeat the identification on the plane of ν and J, for $0 < \nu \leq 1$. The required figure is that phase diagram, with the two line cuts marked on it.

2.3 Spectra of the gapped phase

At one point inside the gapped phase found on the half-filling cut, compute three spectra.

- The charge spectrum measured by scanning tunneling: differential conductance versus bias.
- The spin excitation spectrum that enters muon spin relaxation or nuclear spin-lattice relaxation, as a function of excitation energy.
- The resistivity, or the conductivity, as a function of temperature.

If a spectrum is not a direct ground-state expectation value, state the approximation that produces it. Give every energy in units of t_1.